



Chemistry: CYL1010

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Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, **Bioinorganic Chemistry**

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

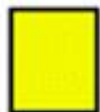
Bioinorganic chemistry

Bioinorganic chemistry is the study of the 'inorganic' elements as they are utilized in biology. The main focus is on metal ions, where we are interested in their interaction with biological ligands and the important chemical properties, they are able to exhibit and impart to an organism. These properties include ligand binding, catalysis, signalling, regulation, sensing, defence, and structural support.

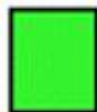
- Study of the structural and functional aspects of metal bound species, such as proteins and nucleic acids in biological systems
- Metal ion transport and storage
- Metallohydrolase enzymes (peptidases)
- Metal-containing electron transfer proteins
- Oxygen transport and activation proteins
- Oxidation and hydroxylation (oxidases)
- Hydrogenases and transferases
- Enzymes involved in nitrogen metabolism pathways

Periodic Table Relevant to Life

1 H 1.0079 Hydrogen																	2 He 4.0026 Helium		
3 Li 6.941 Lithium	4 Be 9.0122 Beryllium													5 B 10.811 Boron	6 C 12.011 Carbon	7 N 14.007 Nitrogen	8 O 15.999 Oxygen	9 F 18.998 Fluorine	10 Ne 20.180 Neon
11 Na 22.990 Sodium	12 Mg 24.305 Magnesium													13 Al 26.982 Aluminum	14 Si 28.086 Silicon	15 P 30.974 Phosphorus	16 S 32.065 Sulfur	17 Cl 35.453 Chlorine	18 Ar 39.948 Argon
19 K 39.098 Potassium	20 Ca 40.078 Calcium	21 Sc 44.956 Scandium	22 Ti 47.867 Titanium	23 V 50.942 Vanadium	24 Cr 51.996 Chromium	25 Mn 54.938 Manganese	26 Fe 55.845 Iron	27 Co 58.933 Cobalt	28 Ni 58.693 Nickel	29 Cu 63.546 Copper	30 Zn 65.38 Zinc	31 Ga 69.723 Gallium	32 Ge 72.631 Germanium	33 As 74.922 Arsenic	34 Se 78.96 Selenium	35 Br 79.904 Bromine	36 Kr 83.80 Krypton		
37 Rb 85.468 Rubidium	38 Sr 87.62 Strontium	39 Y 88.906 Yttrium	40 Zr 91.224 Zirconium	41 Nb 92.906 Niobium	42 Mo 95.94 Molybdenum	43 Tc 98 Technetium	44 Ru 101.07 Ruthenium	45 Rh 102.91 Rhodium	46 Pd 106.42 Palladium	47 Ag 107.87 Silver	48 Cd 112.41 Cadmium	49 In 114.82 Indium	50 Sn 118.71 Tin	51 Sb 121.76 Antimony	52 Te 127.60 Tellurium	53 I 126.90 Iodine	54 Xe 131.29 Xenon		
55 Cs 132.91 Cesium	56 Ba 137.33 Barium	57 - 71 La - Lu	72 Hf 178.49 Hafnium	73 Ta 180.95 Tantalum	74 W 183.84 Tungsten	75 Re 186.21 Rhenium	76 Os 190.23 Osmium	77 Ir 192.22 Iridium	78 Pt 195.08 Platinum	79 Au 196.97 Gold	80 Hg 200.59 Mercury	81 Tl 204.38 Thallium	82 Pb 207.2 Lead	83 Bi 208.98 Bismuth	84 Po 209 Polonium	85 At 210 Astatine	86 Rn 222 Radon		
87 Fr 223 Francium	88 Ra 226 Radium	89 Ac 227 Actinide	90 Th 232.04 Thorium	91 Pa 231.04 Protactinium	92 U 238.03 Uranium														



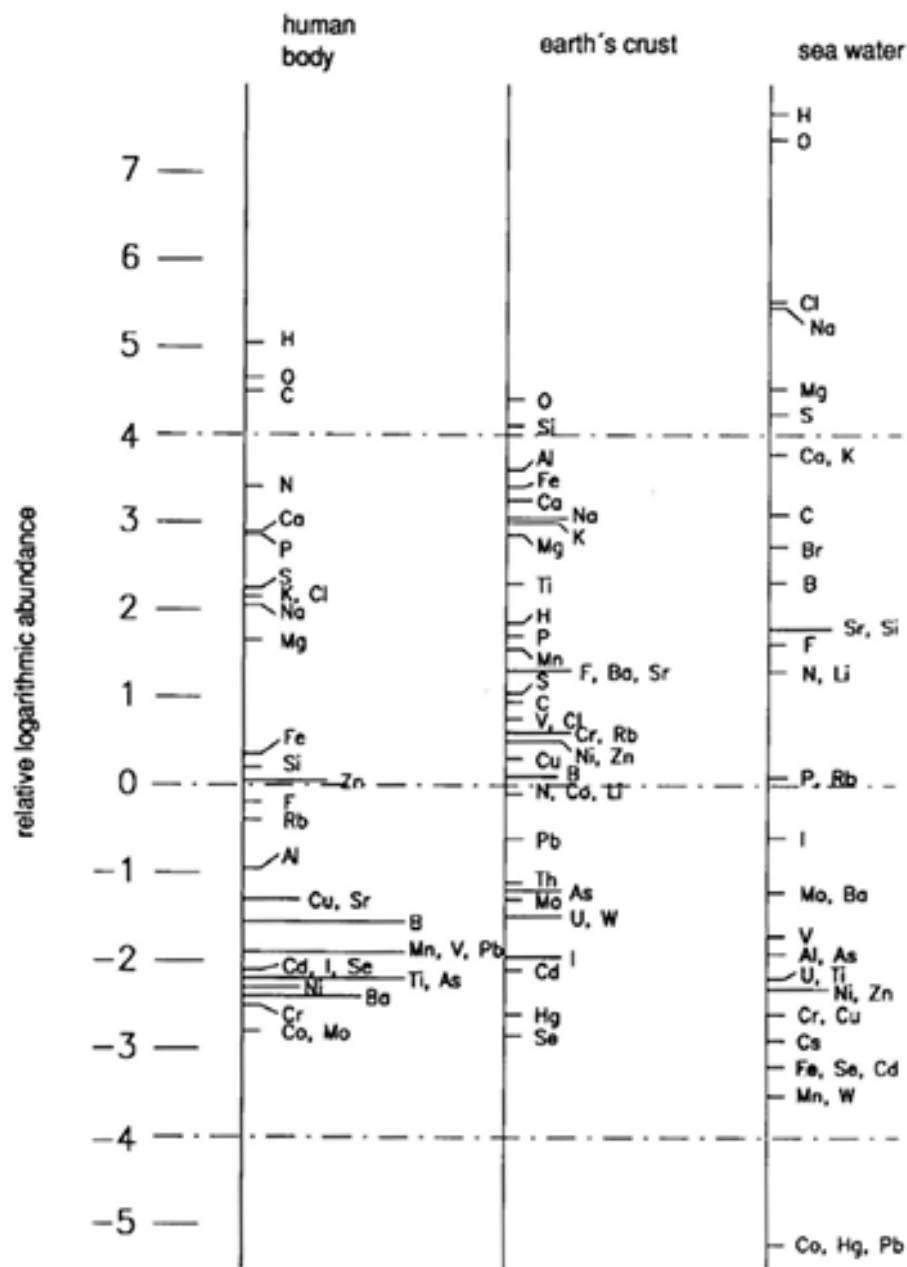
Bulk biological elements



Trace elements believed to be essential for bacteria, plants or animals



Possibly essential trace elements for some species



Element	Sea Water (M) x 10 ⁻⁸	Human Plasma (M) x 10 ⁻⁸
Fe	0.005-2	2230
Zn	8.0	1720
Cu	1.0	1650
Mo	10	1000
V	4.0	17.7
Mn	0.7	10.9
Cr	0.4	5.5
Ni	0.5	4.4
Co	0.7	0.0025

Bertini, I.; Gray, H. B.; Lippard, S. J.; Valentine, J. S.
 Bioinorganic Chemistry; University Science Books:
 Sausalito, CA, 1994.

How nature has chosen these elements?

Criteria for the **selection** of elements

Elemental abundance is not ONLY the determining factor

- Solubility of the element
- Charge type/Oxidation state
- Ionic Radius
- Ligating atoms
- Preferential coordination geometry
- Spin-pairing stabilization
- Kinetic reactivity and other controls
- Thermodynamic aspects
- Chemical reactivity

➤ **Structure**

- Bone and teeth
- Cell membranes
- DNA and RNA structure
- Protein, including enzyme conformation

➤ **Electron transfer (Redox reactions)**

- Fe, Cu, Mn, Mo, Ni, Co

➤ **Metabolism (Degradation of organic molecules)**

➤ **Activation of small molecules (O₂, CO₂)**

Bulk or Constituent Elements:

H, O, C, N, Ca, P, Na, K, S, Cl

Trace Elements:

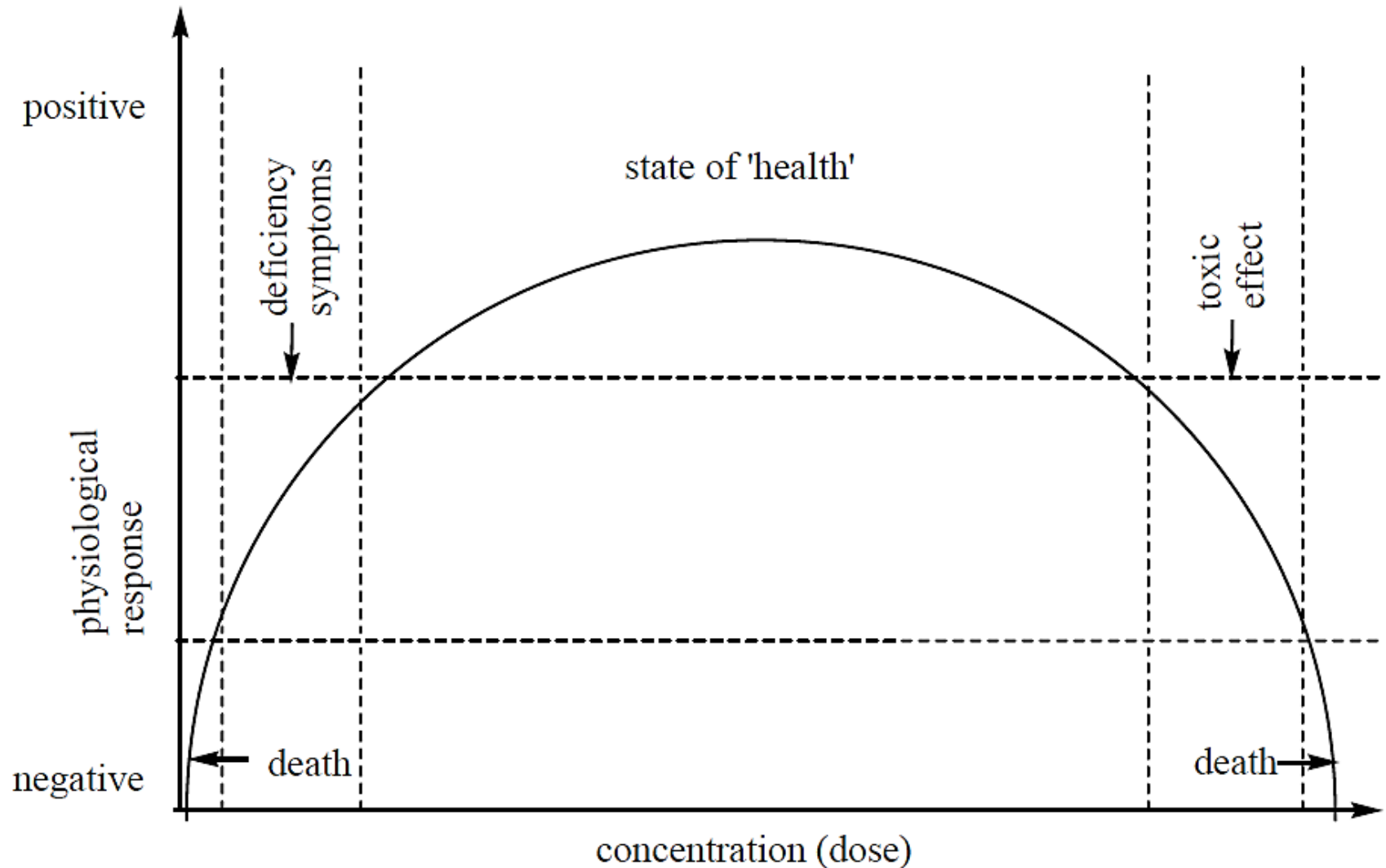
Mg, Si, F, Fe, Zn, B, Rb, Sr, Br, Cu

Ultra Micro Trace Elements:

V, Li, Se, Mn, Ba, Ge, As, Ni, Mo, Cd, I, Sn, Cr, Pb, Co

<i>Metal</i>	<i>Function</i>	<i>Typical deficiency symptoms</i>
Sodium	Charge carrier; osmotic balance	
Potassium	Charge carrier; osmotic balance	
Magnesium	Structure; hydrolase; isomerase	muscle cramps
Calcium	Structure; trigger; charge carrier	retarded skeletal growth
Vanadium	Nitrogen fixation; oxidase	
Chromium	Unknown, possible involvement in glucose tolerance	diabetes symptoms
Molybdenum	Nitrogen fixation; oxidase; oxo transfer	retardation of cellular growth; propensity for caries
Tungsten	Dehydrogenase	
Manganese	Photosynthesis; oxidase; structure	infertility; impaired skeletal growth
Iron	Oxidase; dioxygen transport and storage; electron transfer; nitrogen fixation	anemia; disorders of the immune system
Cobalt	Oxidase; alkyl group transfer	pernicious anemia
Nickel	Hydrogenase; hydrolase	growth depression; dermatitis
Copper	Oxidase; dioxygen transport; electron transfer	artery weakness; liver disorders; secondary anemia
Zinc	Structure; hydrolase	skin damage; stunted growth; retarded sexual maturation

Essential Element Dosage and Physiological Response



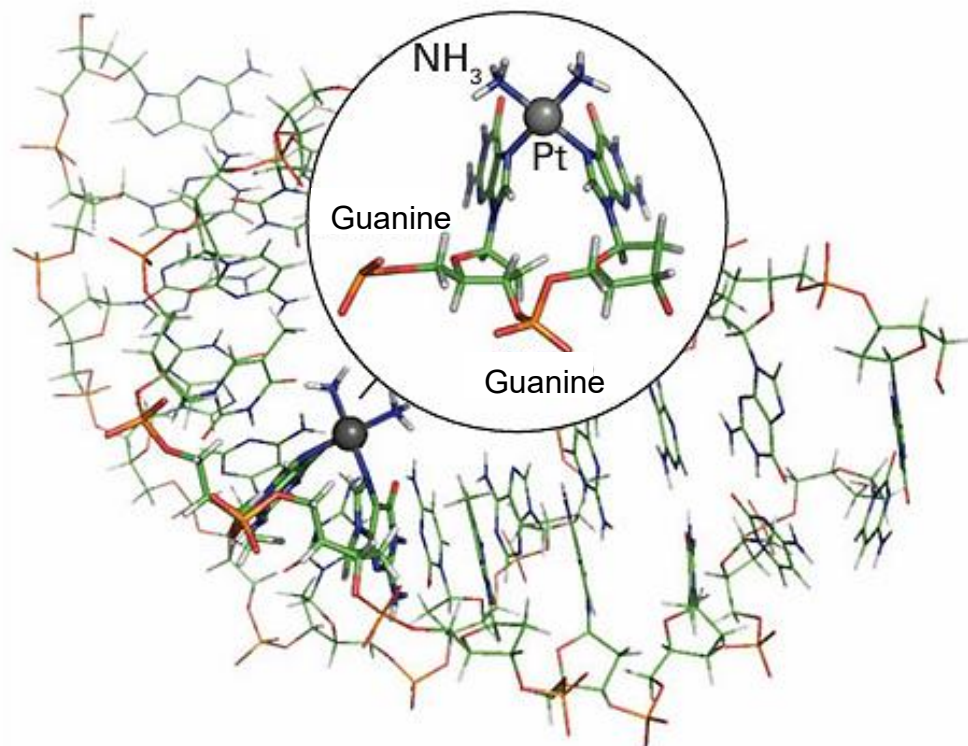
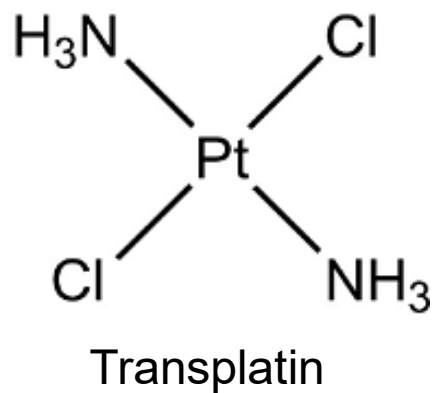
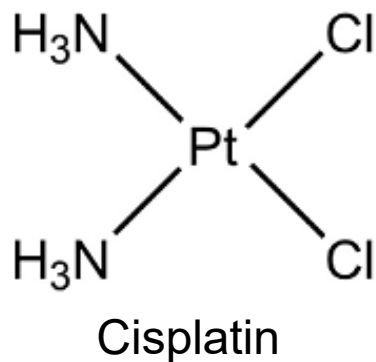
Why does biology utilize transition metals?

- Transition Metals are extremely good catalytic active sites in enzymes, because they:
 - are stable in a variety of geometries and CN
 - have multiple coordination sites
 - Are stable in a variety of oxidation states
 - Are able to change the reactivity of ligands
 - Have “weak” coordinate bonds (where needed)
 - Are capable of stabilizing intermediates

Why does it matter what ligands are attached to the metal ?

- Tune Redox Properties
- Assist in stabilization of multiple oxidation states of transition metals
- Lability/Stability

cis- and *trans*- Diamminedichloroplatinum (II)



Cis-Platin is a highly effective chemotherapy drug for cancers due to its ability of cross-linking with DNA and inhibiting to copy the DNA by enzymes

Terms and Definitions in Bioinorganic Chemistry

Active center: Location in an enzyme where the specific reaction takes place.

Apo-enzyme: An enzyme that lacks its metal center.

Charge-transfer complex: An aggregate of two or more molecules in which charge is transferred from a donor to an acceptor.

Cluster: Metal centers grouped close together which can have direct metal bonding or through a bridging ligand, e.g. ferredoxin

Cofactor: An organic molecule or ion (usually a metal ion) that is required by an enzyme for its activity. It may be attached either loosely (coenzyme) or tightly (prosthetic group).

Cooperativity: The phenomenon that binding of an effector molecule to a biological system either enhances or diminishes the binding of a successive molecules, e.g. hemoglobin.

Terms and Definitions in Bioinorganic Chemistry

Entatic state: A state of an atom or group which has its geometric or electronic condition adapted for function. Derived from entasis (Greek) meaning tension.

Enzyme: A macromolecule that functions as a biocatalyst by increasing the reaction rate.

Heme: A near-planar coordination complex obtained from iron and dianionic porphyrin.

Hemoglobin: A dioxygen-carrying heme protein of red blood cells.

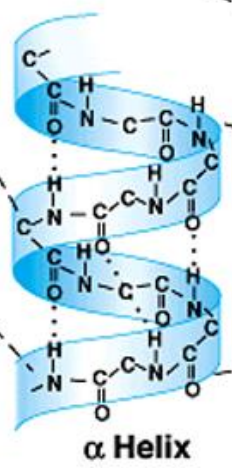
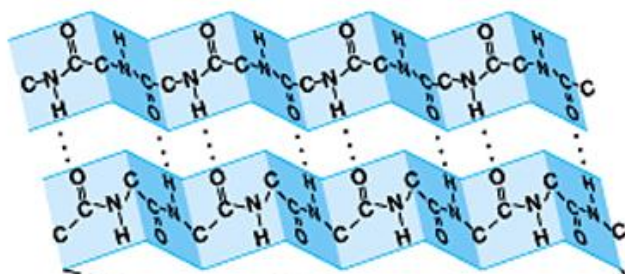
Holoenzyme: An enzyme containing its characteristic prosthetic group(s) and/or metal(s).

Ion channel: Enable ions to flow rapidly through membranes in a thermodynamically downhill direction after an electrical or chemical impulse.

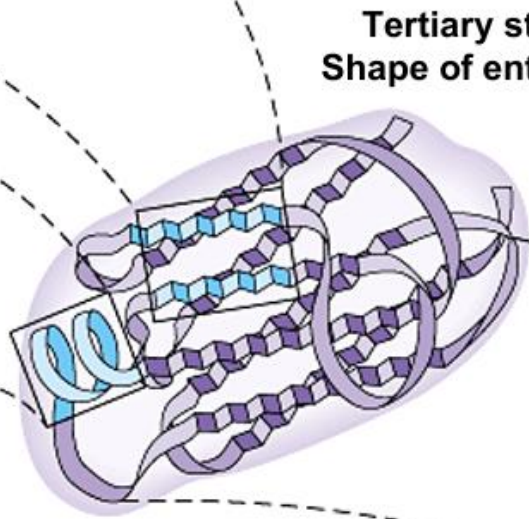
Ionophore: A compound which can carry specific ions through membranes.

Ion pumps: Enable ions to flow through membranes in a thermodynamically uphill direction by the use of an energy source. They open and close upon the binding and subsequent hydrolysis of ATP, usually transporting more than one ion towards the outside or the inside of the membrane.

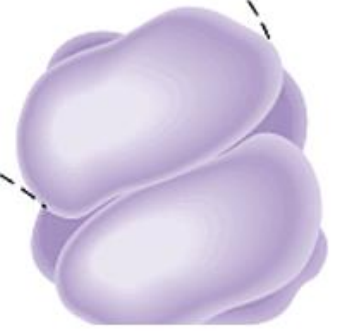
Primary structure:
Sequence of amino acids



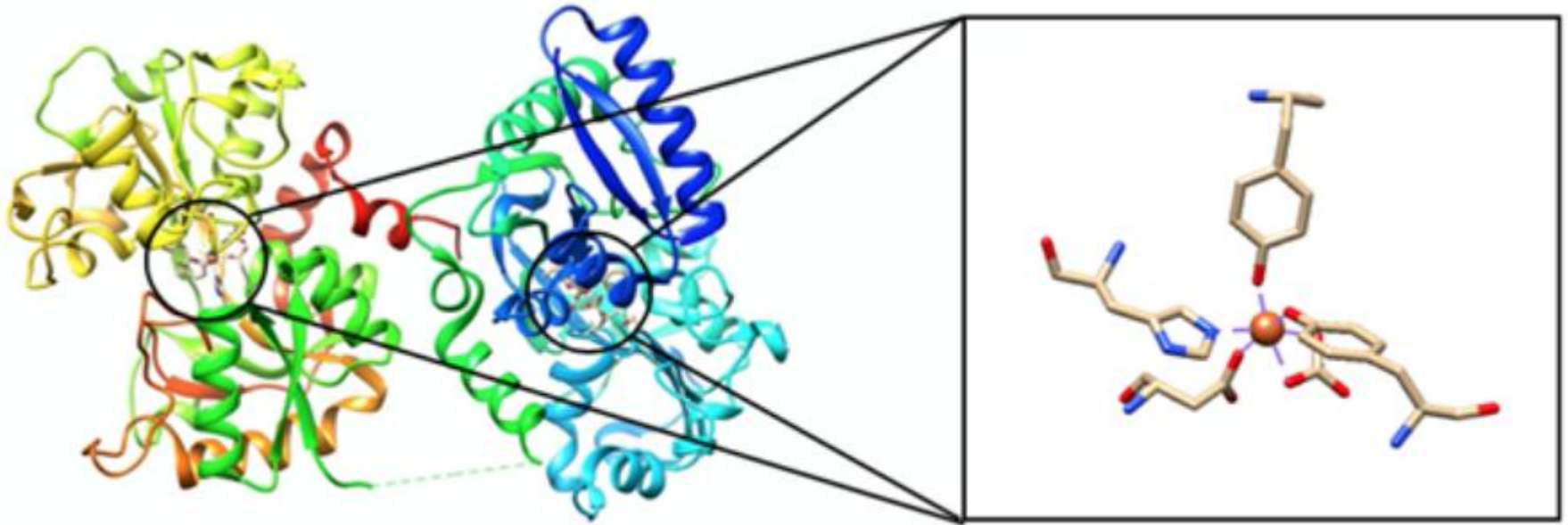
Secondary structure:
Shapes formed within regions of the protein



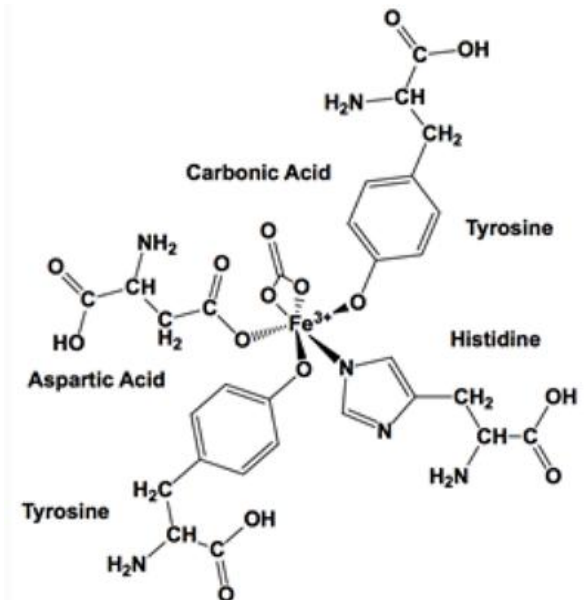
Quaternary structure:
Structures formed by interaction of several subunit



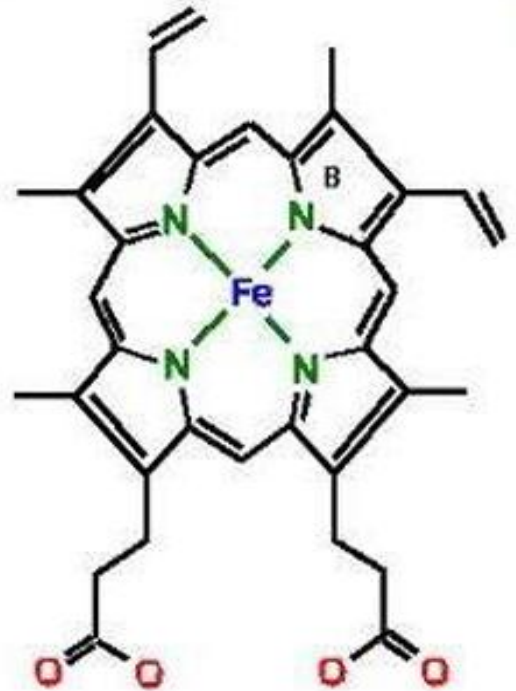
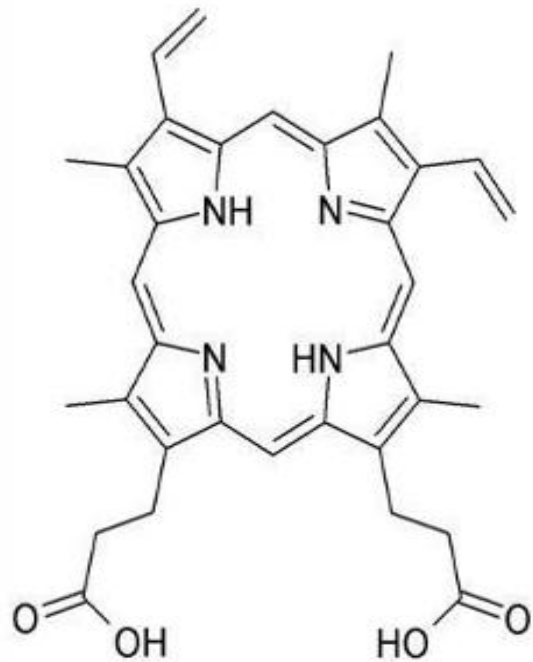
Human Serum Transferrin



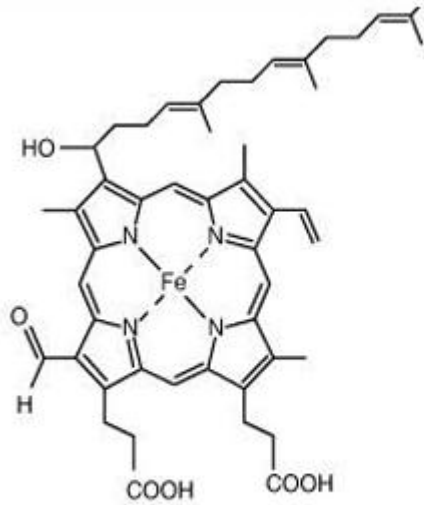
- This protein transports iron ions to various organs of the body
- Picks up iron from the storage protein, viz., ferritin, goes through the blood and delivers at the organ tissue
- During pick up and delivery iron is in +2; and during transport and storage it is in +3



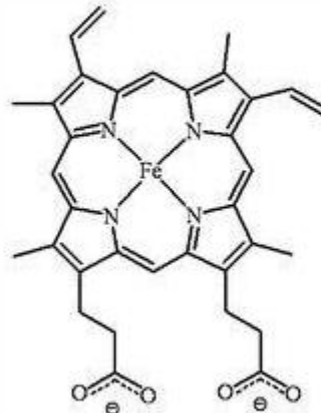
Porphyrin



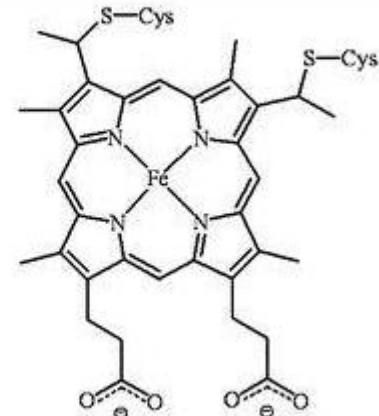
Heme and Haemoglobin



Heme A

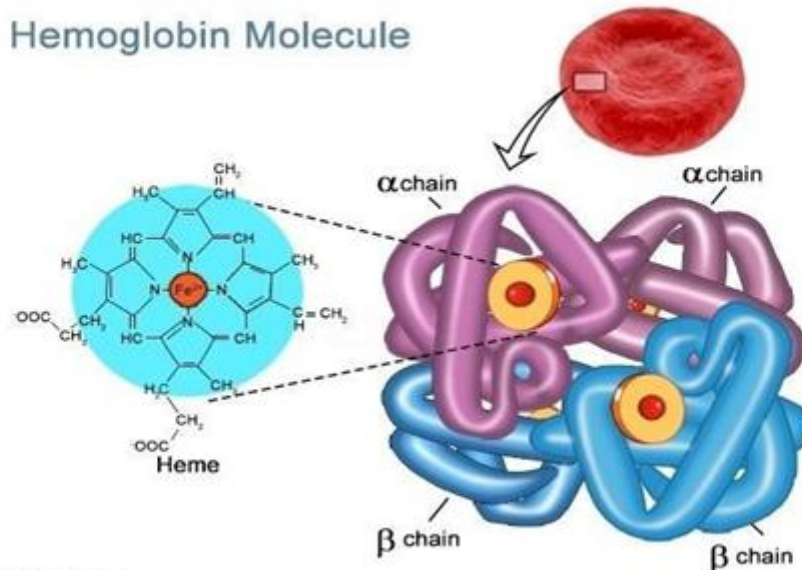


heme *b*

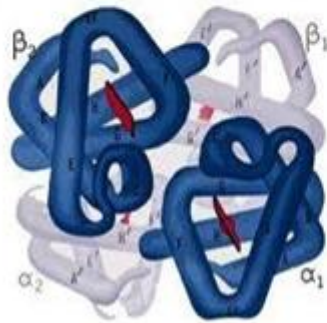


heme *c*

Hemoglobin Molecule



Hemoglobin Hb



Four units of Hb

3 major types of Hb

Hb A (Adult)

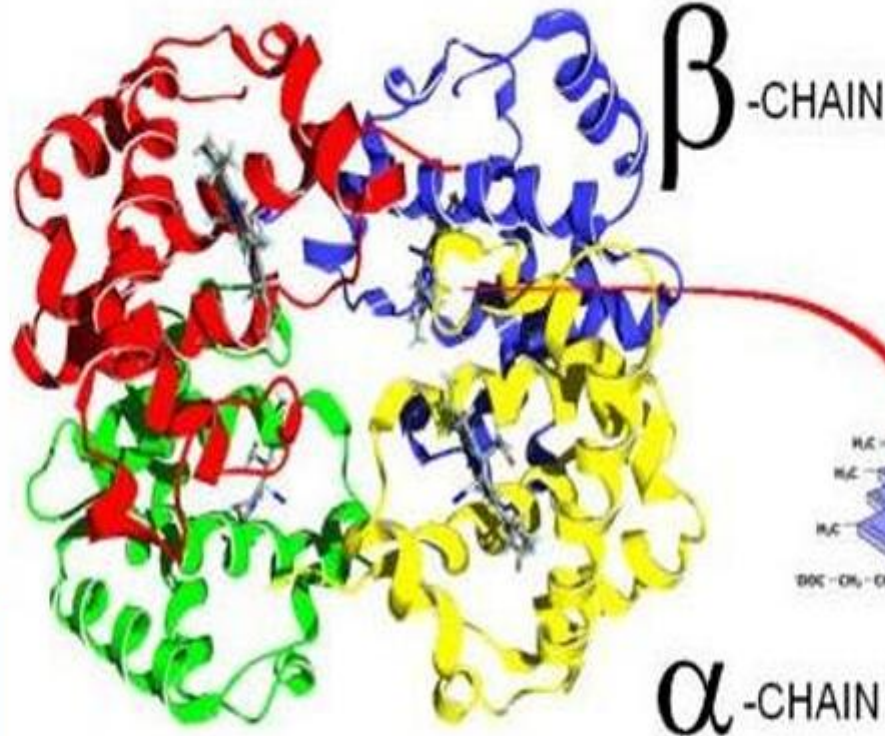
Hb F (Fetal)

Hb S (Sickle cell)

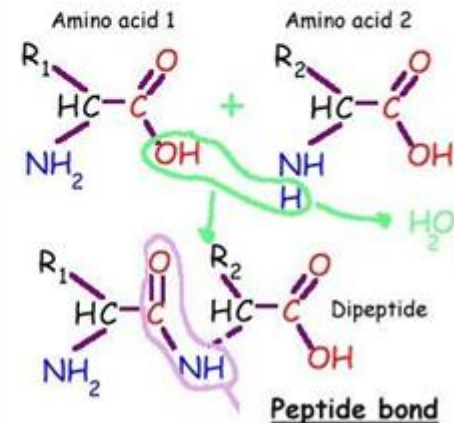
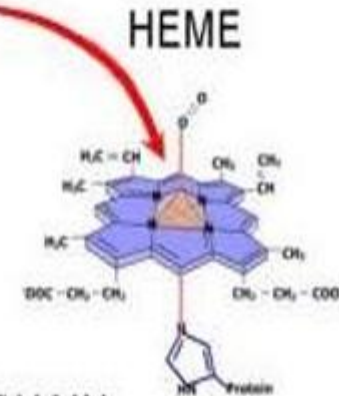
α Chain = 141 amino acid

β Chain = 146 amino acid

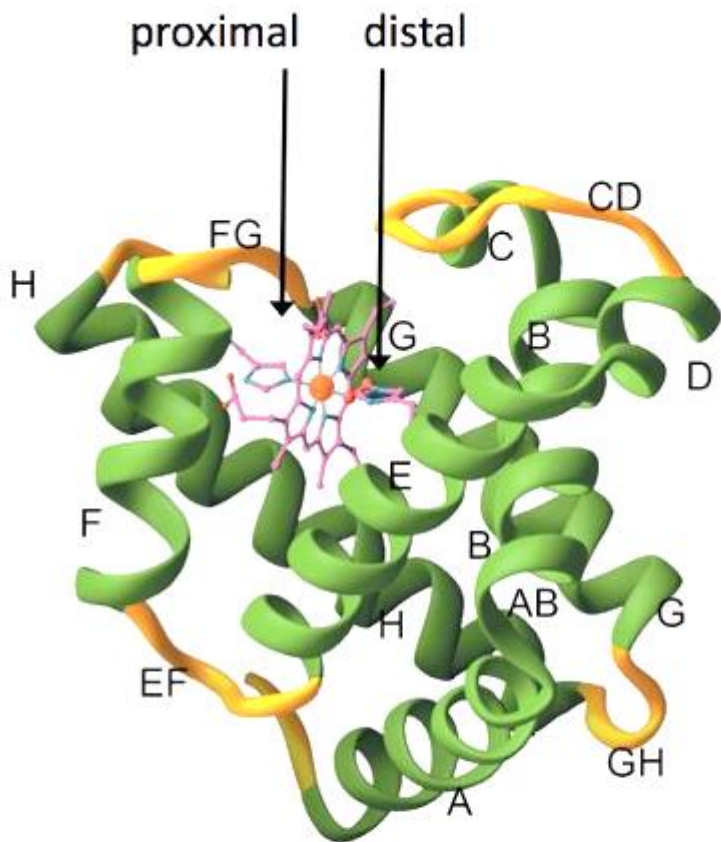
2 α and 2 β peptide chain are interlinked through H bonded interaction ($\text{COO}^- \cdots \text{NH}_3^+$)



Hb is tetrameric protein
M.W = 64500 Dalton



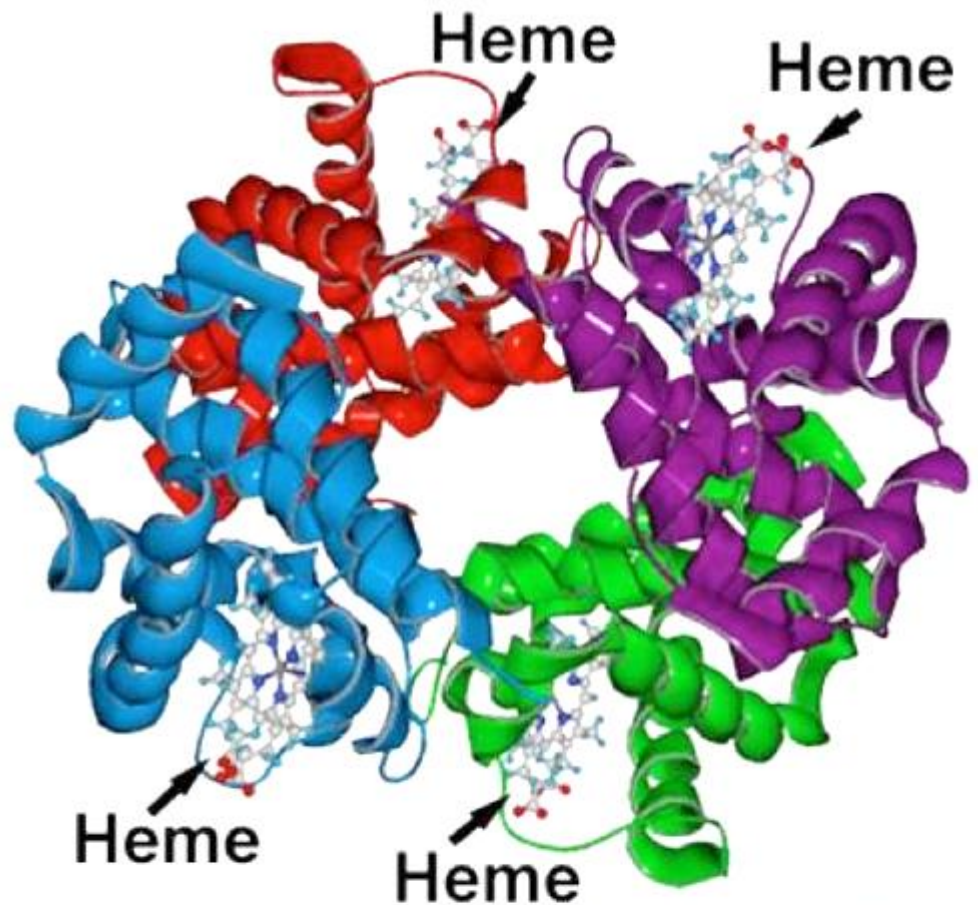
Peptide bond



V&V p 244 Fig. 8-39b Ribbon Structure of Myoglobin

Oxygenated Myoglobin

monomeric



www.animalscience.ucdavis.edu

Hemoglobin

Tetrameric

Myoglobin – O₂ storage; Hemoglobin – O₂ transport

Role of the protein in case of hemoglobin

Binding pocket of O₂ in protein:

Prevent 2-e reduction

Prevent μ -oxo dimer formation

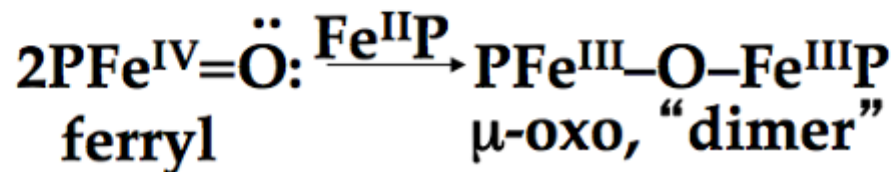
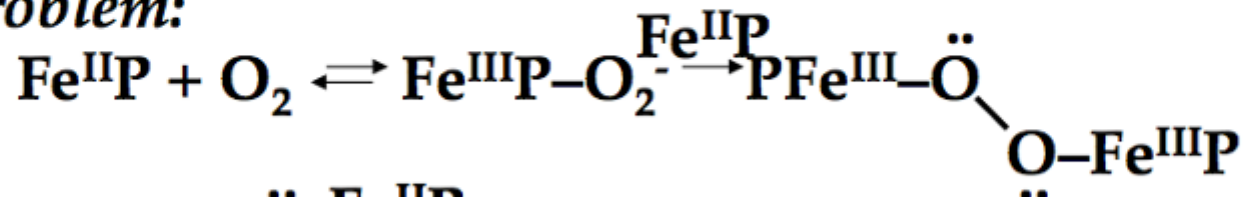
Stabilizing PFe(II)...O₂ complex

Bent O₂ geometry

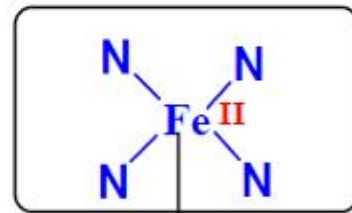
Binding of CO vs. O₂

Thermodynamics vs. Kinetics -- Role of the protein

The problem:



Oxy- and de-oxy forms of Hemoglobin

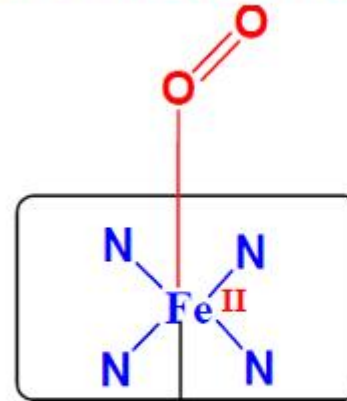


N

(His)

High Spin
paramagnetic
 $t_2g^4 e_g^2$

Deoxyhemoglobin



N

(His)

Low Spin
diamagnetic
 $t_2g^6 e_g^0$

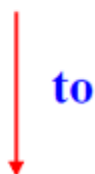
Oxyhemoglobin

Deoxyhemoglobin is the form of hemoglobin without the bound oxygen. The oxyhemoglobin has significantly lower absorption (660 nm) than deoxyhemoglobin (940 nm). This difference is used for measurement of the amount of oxygen in patient's blood by pulse oximeter.

What happens when O_2 binds to Hemoglobin

The size of Fe^{2+} increase by 28% on going from

Low spin (oxyhemoglobin) ($0.61 \text{ \AA}$)

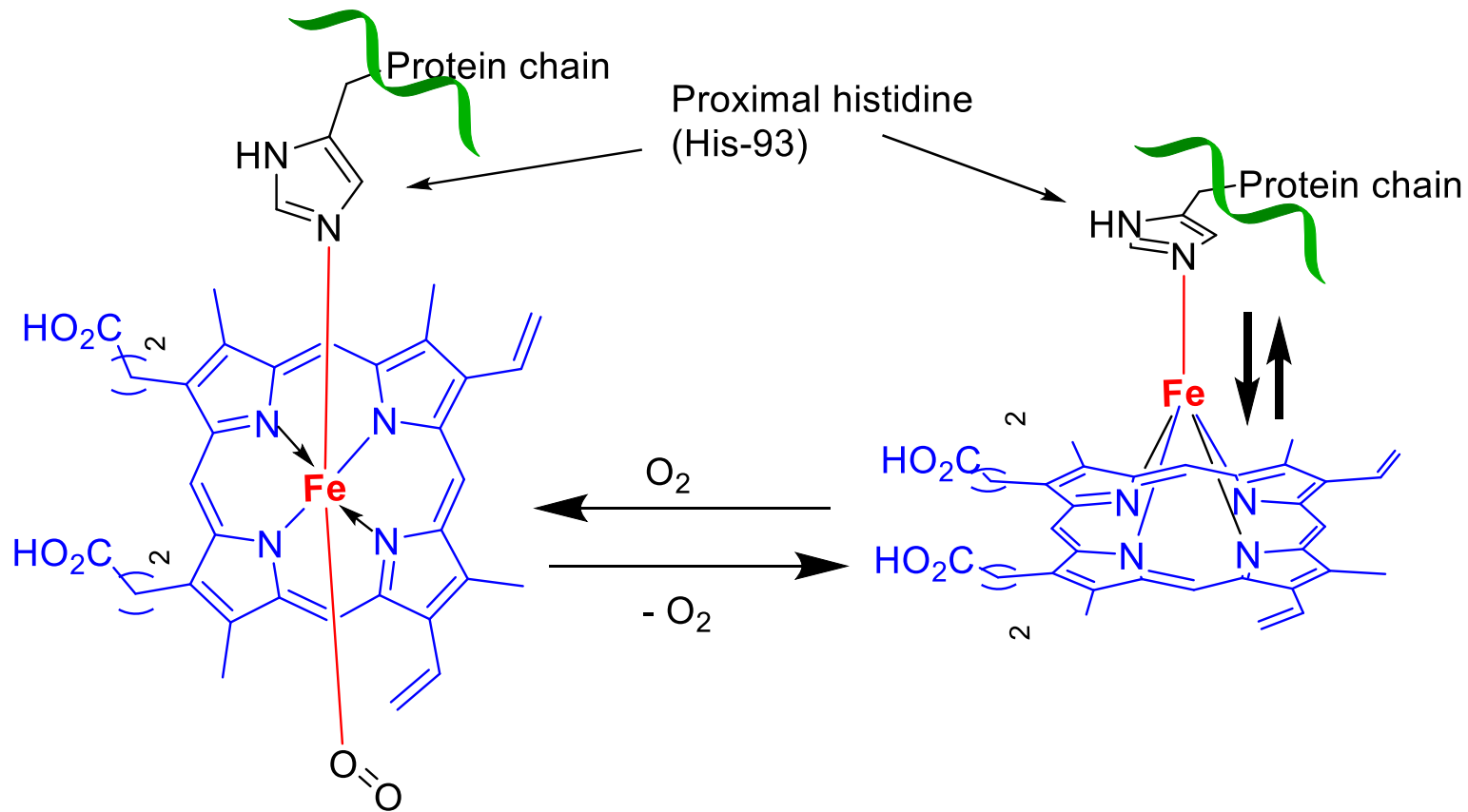


High spin (Deoxyhemoglobin) ($0.78 \text{ \AA}$)

The Fe^{2+} in deoxyhemoglobin is too large to fit in the ring and is situated $(0.7-0.8) \text{ \AA}$ above the ring

Thus, presence of O_2 changes the electronic arrangement of Fe^{2+} and distorts the shape of the complex

The globular protein prevents the irreversible oxidation of $Fe(II)$ to $Fe(III)$



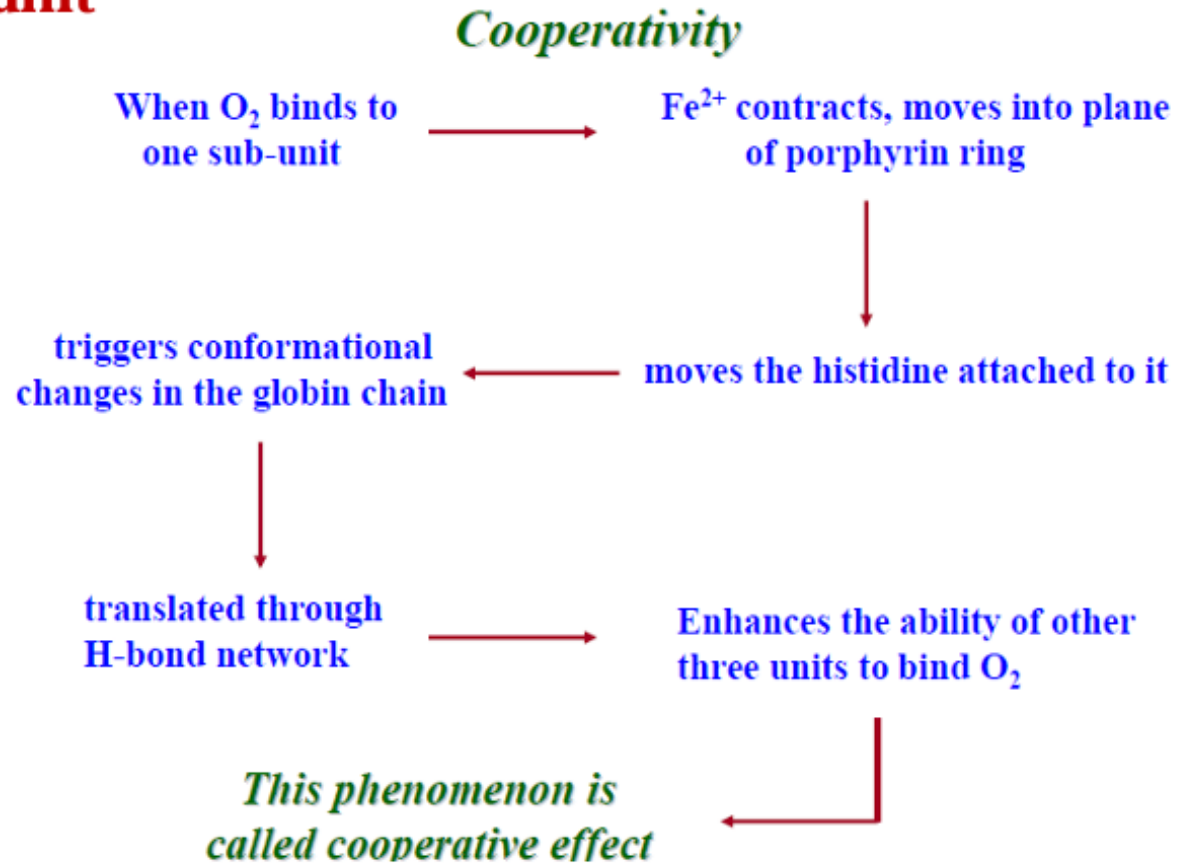
Oxymyoglobin/Oxyhemoglobin

Six coordinated iron atom,
Low spin, in the plane of heme
R-state

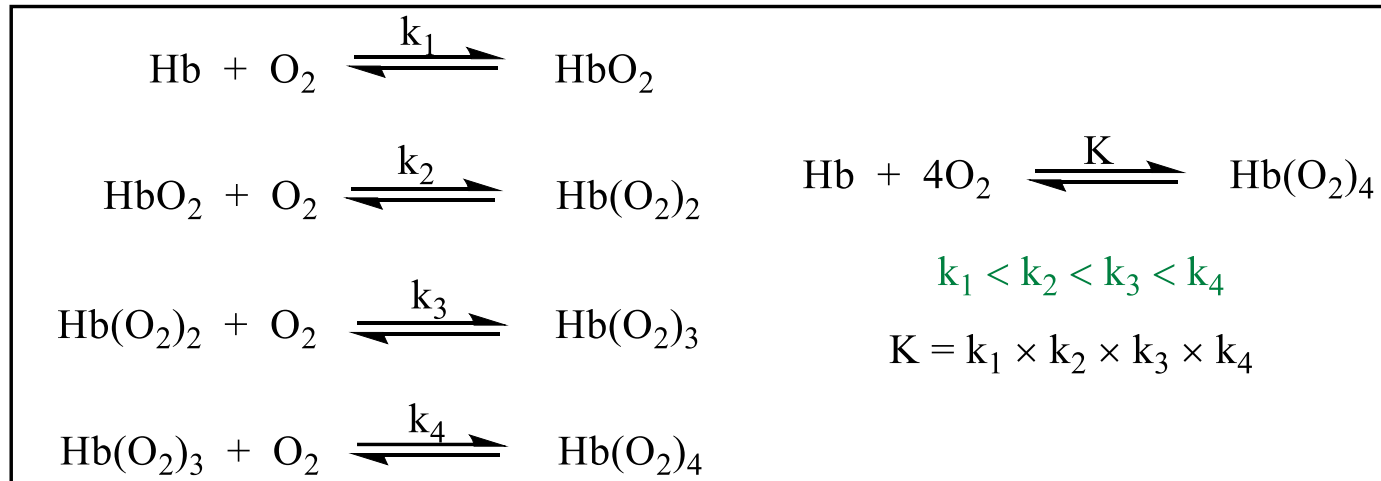
Deoxymyoglobin/deoxyhemoglobin

Five coordinated iron atom,
High spin, out of the plane of heme,
T-state

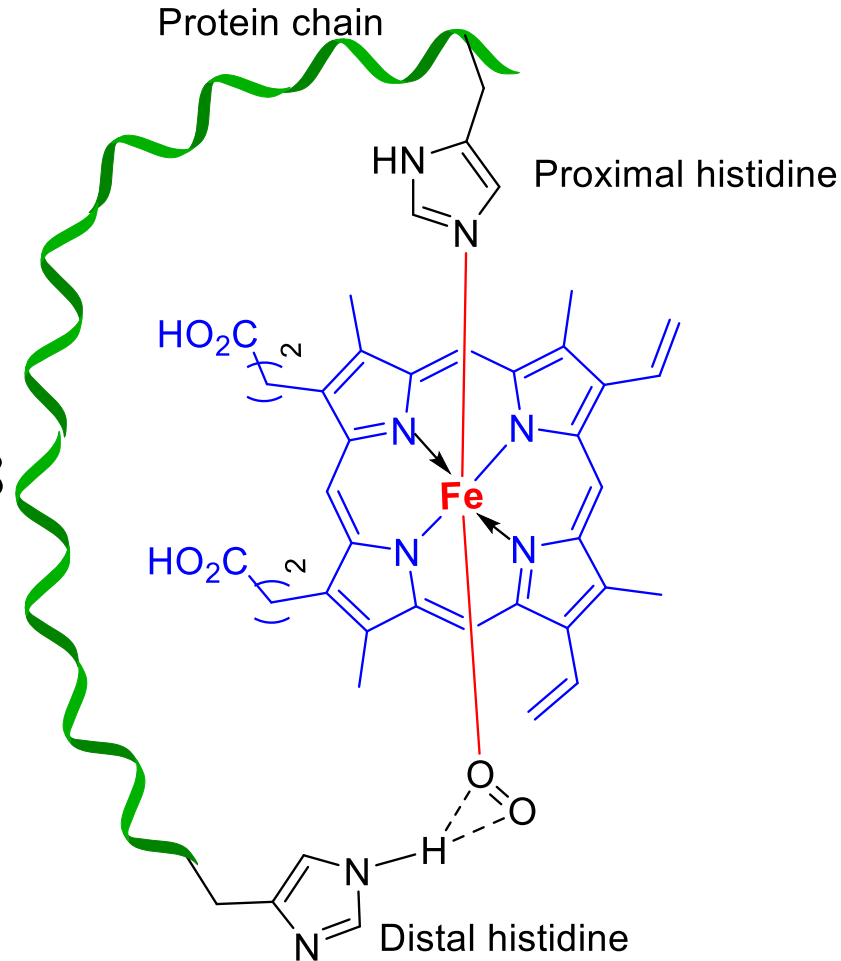
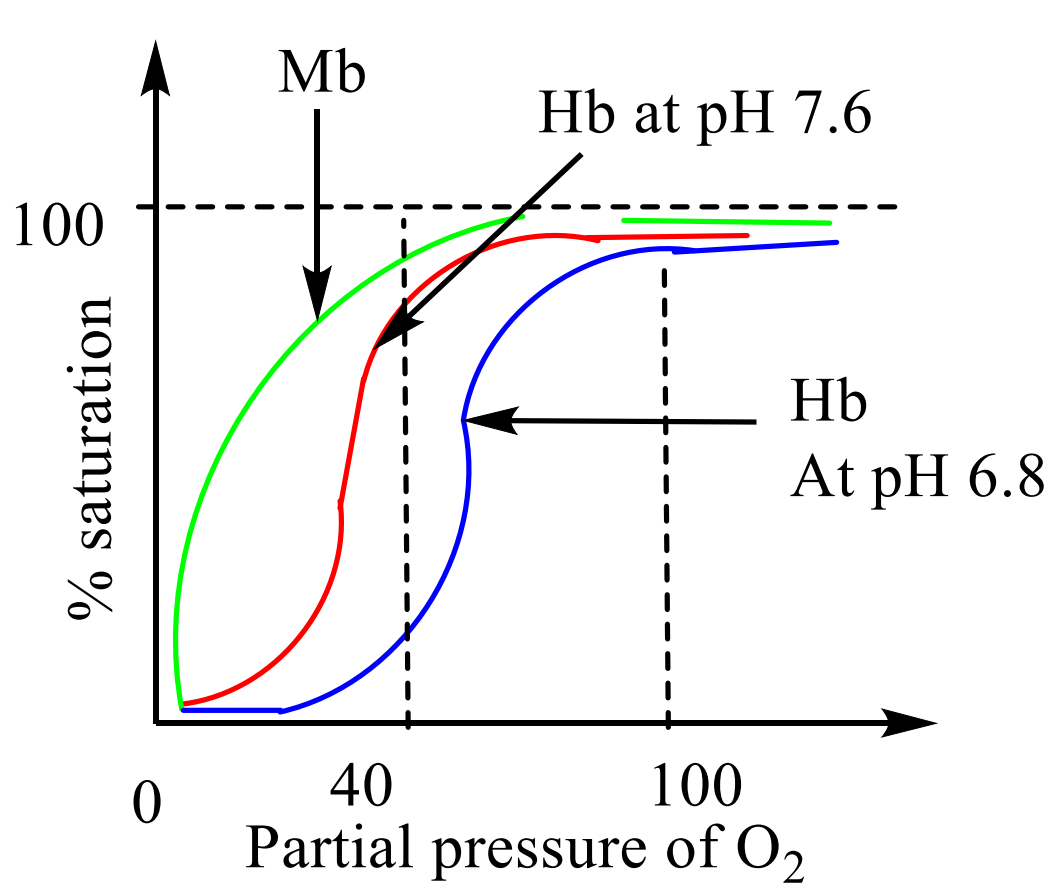
Co-ordination of one O₂ leads to conformational changes in the protein chain leading to facilitate co-ordination of O₂ by other 3-sub-unit



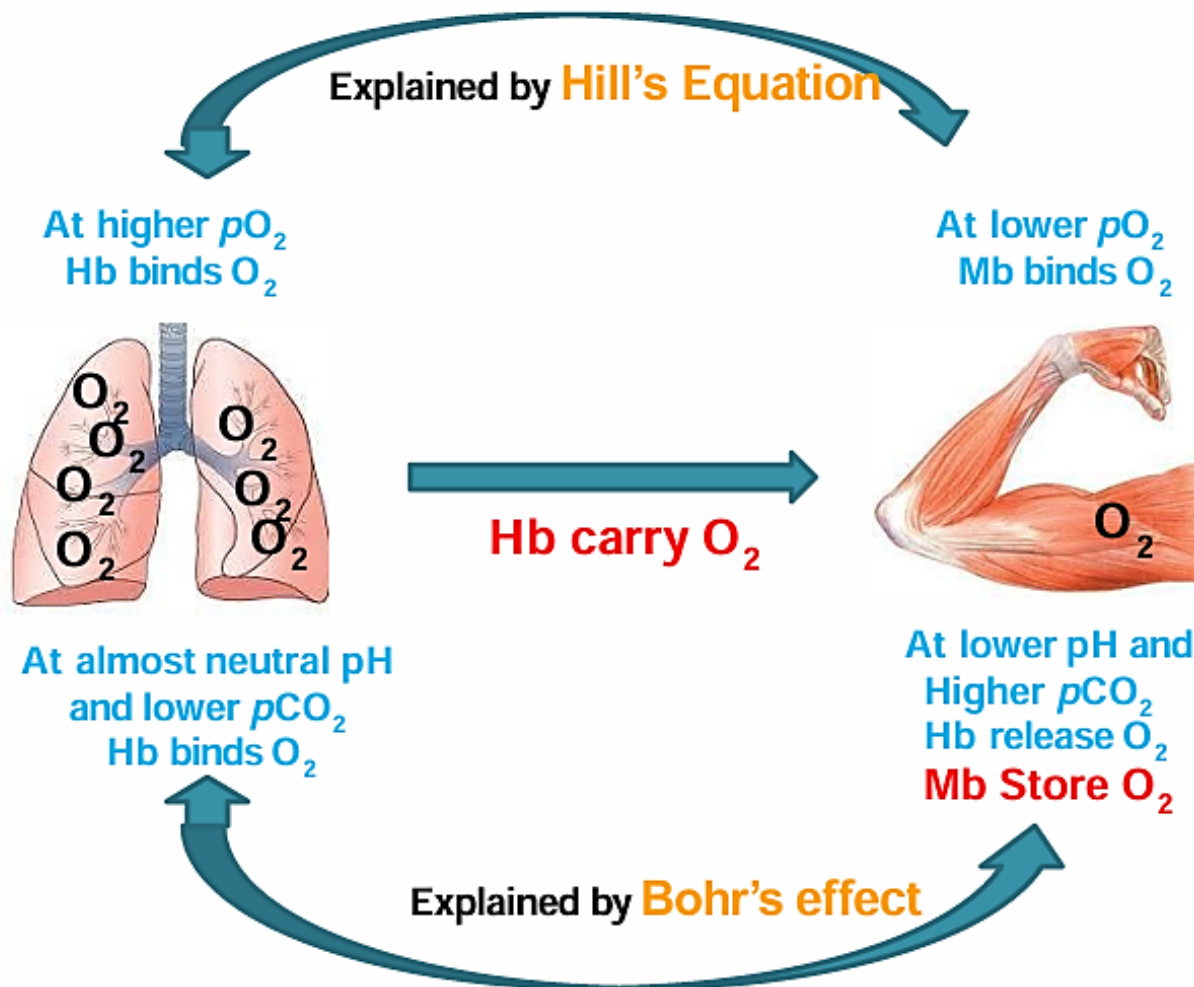
Transport of dioxygen



Transport of dioxygen



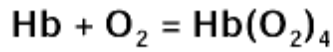
O_2 binding curve for Hb and Mb



Hill Equation

$$\theta = \frac{[L]^n}{K_d + [L]^n}$$

- θ is the fraction of the receptor protein concentration that is bound by the ligand,
- $[L]$ is the total ligand concentration,
- K_d is the apparent dissociation constant derived from the law of mass action,
- n is the Hill coefficient.

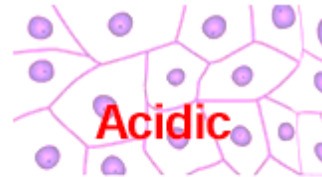


Hb carry O_2

At high O_2 pressure
High O_2 affinity of Hb
Hb release CO_2

Acidic condition

CO_2 transported to lungs

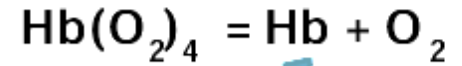


Animal Tissues

Acidic

produce H_2CO_3 ,
 CO_2 , lactic acid

Acidic condition



+ Mb
pH independent

$\text{Mb}(\text{O}_2)$
Store for
respiration

Hb reverse back to lungs with CO_2

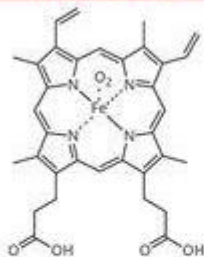
THE CHEMISTRY OF THE DIFFERENT COLOURS OF BLOOD



Red

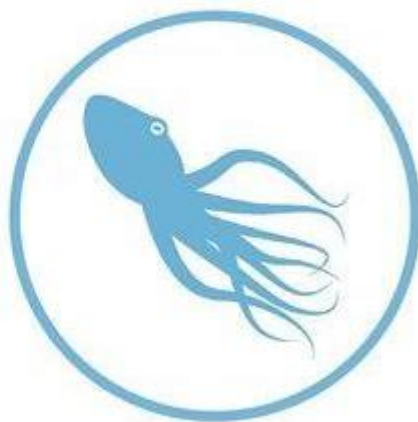
HUMANS AND THE MAJORITY OF
OTHER VERTEBRATES

HAEMOGLOBIN



HAEMOGLOBIN
(oxygenated form)

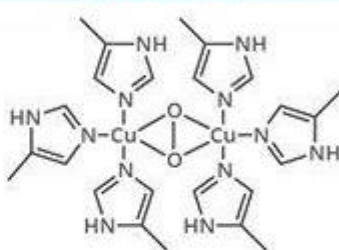
Haemoglobin is a protein found in blood, built up from subunits called 'haems'. These subunits contain iron, and their structure gives blood its red colour when oxygenated. Deoxygenated blood is a deep red colour - not blue!



Blue

SPIDERS, CRUSTACEANS, SOME
MOLLUSCS, OCTOPUSES & SQUID

HAEMOCYANIN



HAEMOCYANIN
(oxygenated form)

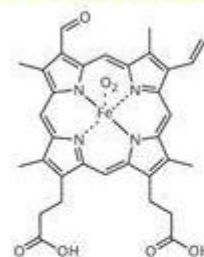
Unlike haemoglobin, which is bound to red blood cells, haemocyanin floats free in the blood. Haemocyanin contains copper instead of iron. When deoxygenated, the blood is colourless, but when oxygenated, it gives a blue colouration.



Green

SOME SEGMENTED WORMS, SOME
LEECHES, & SOME MARINE WORMS

CHLOROCRUORIN



CHLOROCRUORIN
(deoxygenated form)

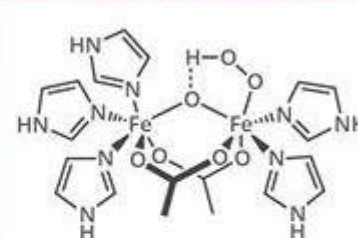
Chemically similar to haemoglobin; the blood of some species contains both haemoglobin & chlorocruorin. Light green when deoxygenated, it is green when oxygenated, although when more concentrated it appears light red.



Violet

MARINE WORMS INCLUDING PEANUT
WORMS, PENIS WORMS & BRACHIOPODS

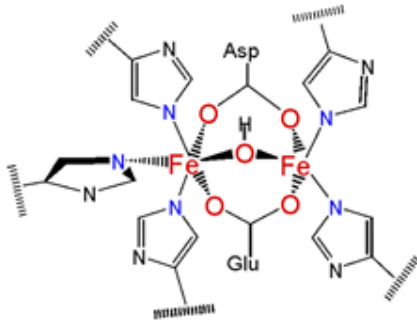
HAEMORYTHRIN



HAEMORYTHRIN
(oxygenated form)

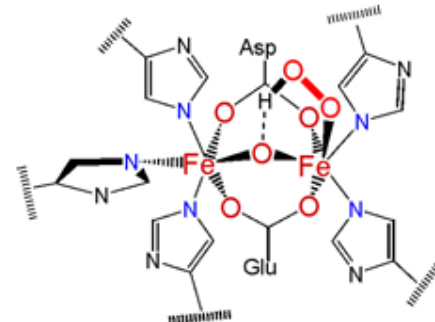
Haemerythrin is only 1/4 as efficient at oxygen transport when compared to haemoglobin. In the deoxygenated state, haemerythrin is colourless, but it imparts a violet-pink colour when oxygenated.





Deoxyhemerythrin, h.s. Fe(II), CN = 5, 6

Hemerythrin

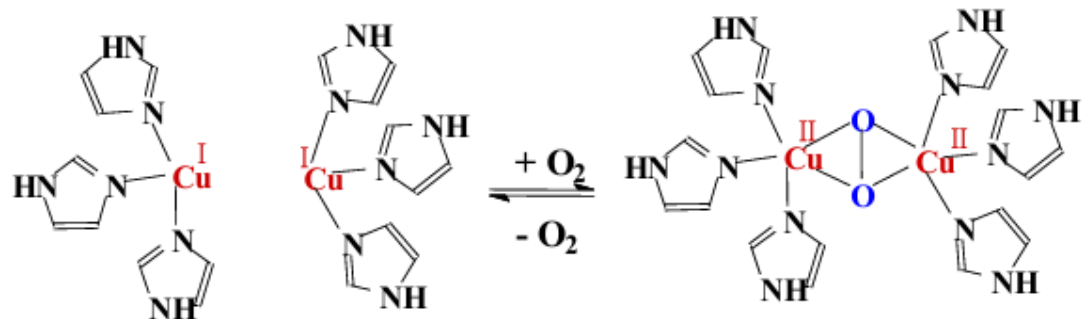


Oxyhemerythrin, Fe(III) with superexchange

- Non-porphinoid metal proteins for reversible O_2 uptake (spiders, etc.)
- Attachment of 3O_2 at Fe with CN = 5,
- Oxidation of both Fe (Fe(III)), reduction of 3O_2 to peroxide (oxo-bridge via H-transport to peroxide)

Hemocyanin

- Proteins with two Cu-centers
- Reversible oxidative addition:
 $O_2 \rightarrow O_2^{2-}$, $Cu^+ \rightarrow Cu^{2+}$
- Side-on bridging coordination

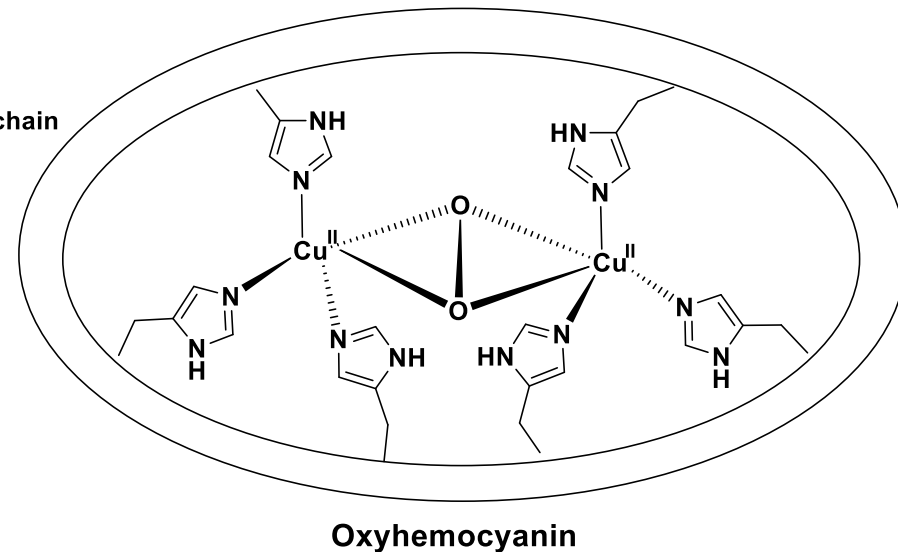
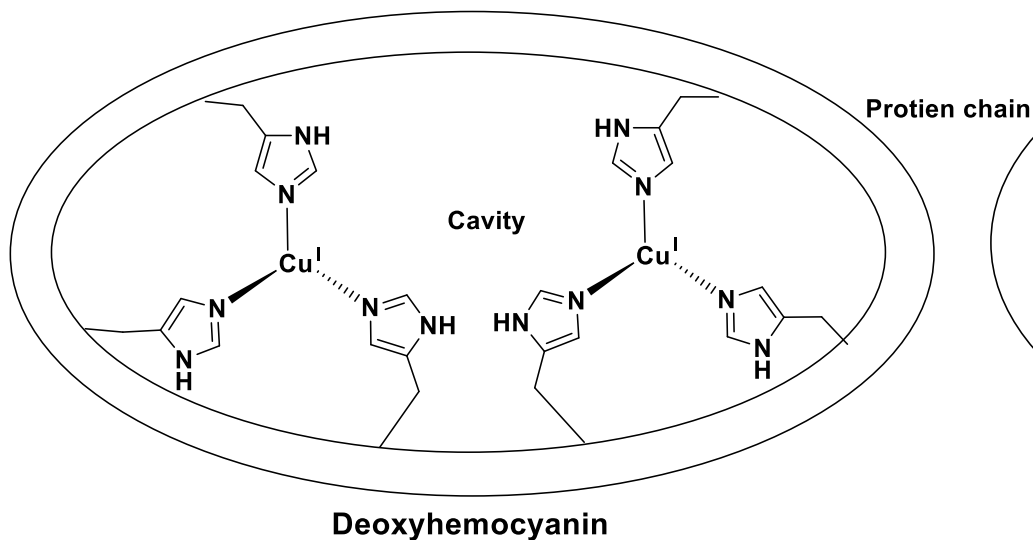


Oxygen binding in hemerythrin

- Hemerythrin holds the O_2 as a hydroperoxide
- Colorless when deoxygenated, but turn a violet-pink in the oxygenated state
- Iron atoms are bound to the protein through the carboxylate side chains of a glutamate and aspartates as well as through five histidine residues.

Hemocyanin

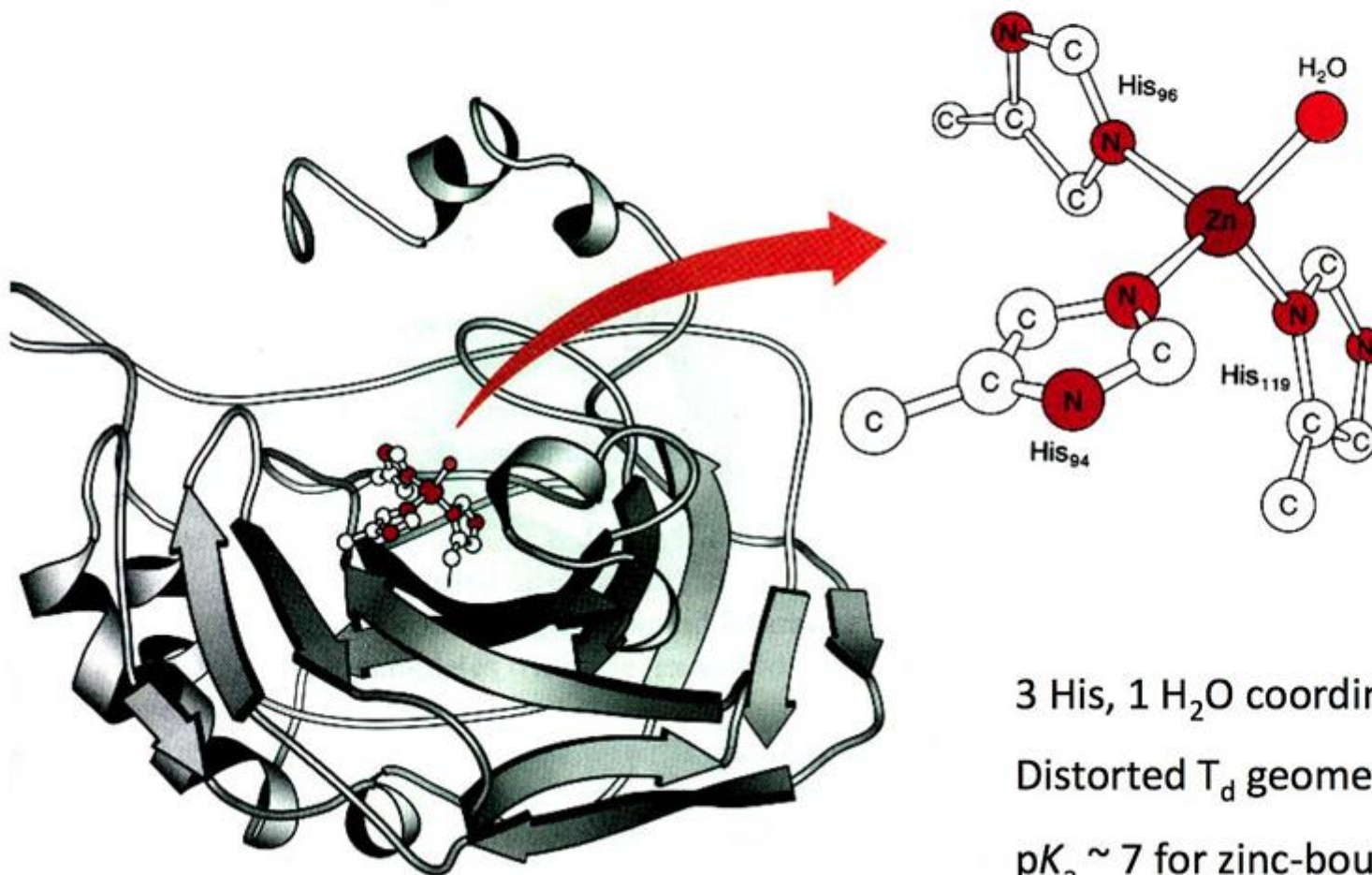
- Found in some Mollusca and Arthropoda species
- Deoxy form is colourless and oxy form is blue
- The oxygen binding center consists of two copper atoms, which exhibits +1 oxidation state in deoxy-form and +2 oxidation state in oxy-form
- Hemocyanin binds dioxygen in peroxide form and each oxygen is bonded with both copper centers.



Carbonic Anhydrase: Mononuclear Zinc Lyase



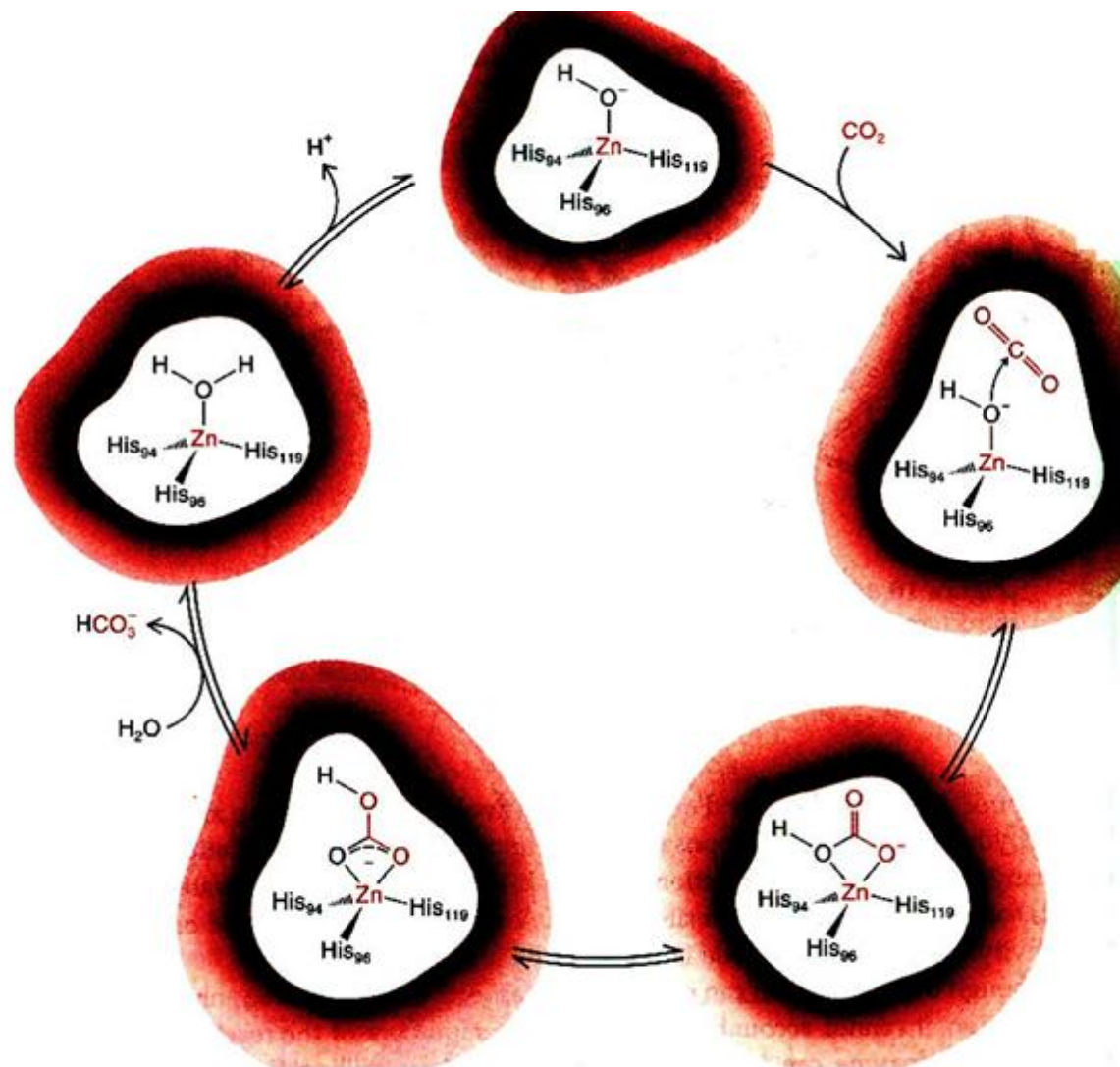
Rates: no enzyme, 10^{-2} s^{-1} ; enzyme, 10^6 s^{-1}



3 His, 1 H₂O coordinate Zn(II)

Distorted T_d geometry

pK_a ~ 7 for zinc-bound H₂O

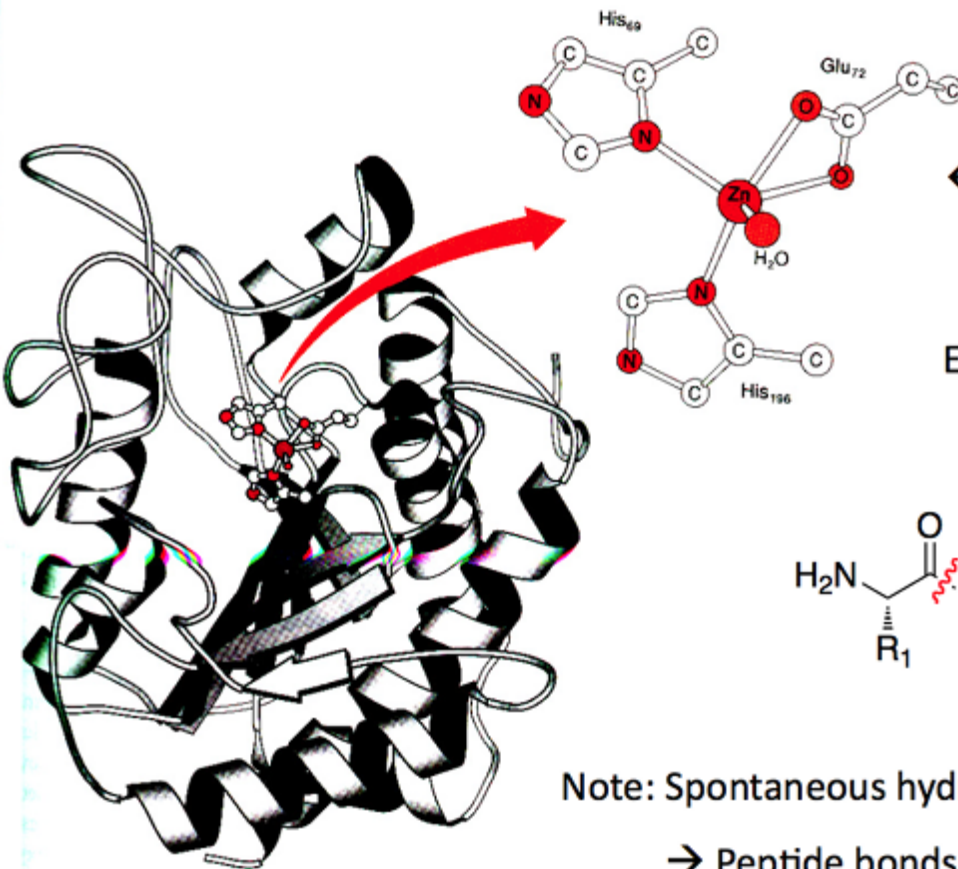


Carboxypeptidase A

Carboxypeptidase A is a **metalloprotease** that hydrolyzes C-terminal peptide bonds

35 kDa, **pentacoordinate zinc** in the absence of substrate or inhibitor

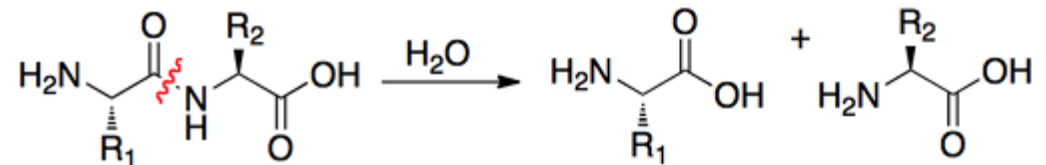
Glu₇₇ undergoes **carboxylate shift** with substrate/inhibitor coordination → bi- to monodentate



← Bidentate Glu₇₇

Ex. Hydrolysis of 2-mer peptide

(aromatic & branched sidechains preferred)



Note: Spontaneous hydrolysis at pH 7 → $k = 3 \times 10^{-9} \text{ sec}^{-1}$

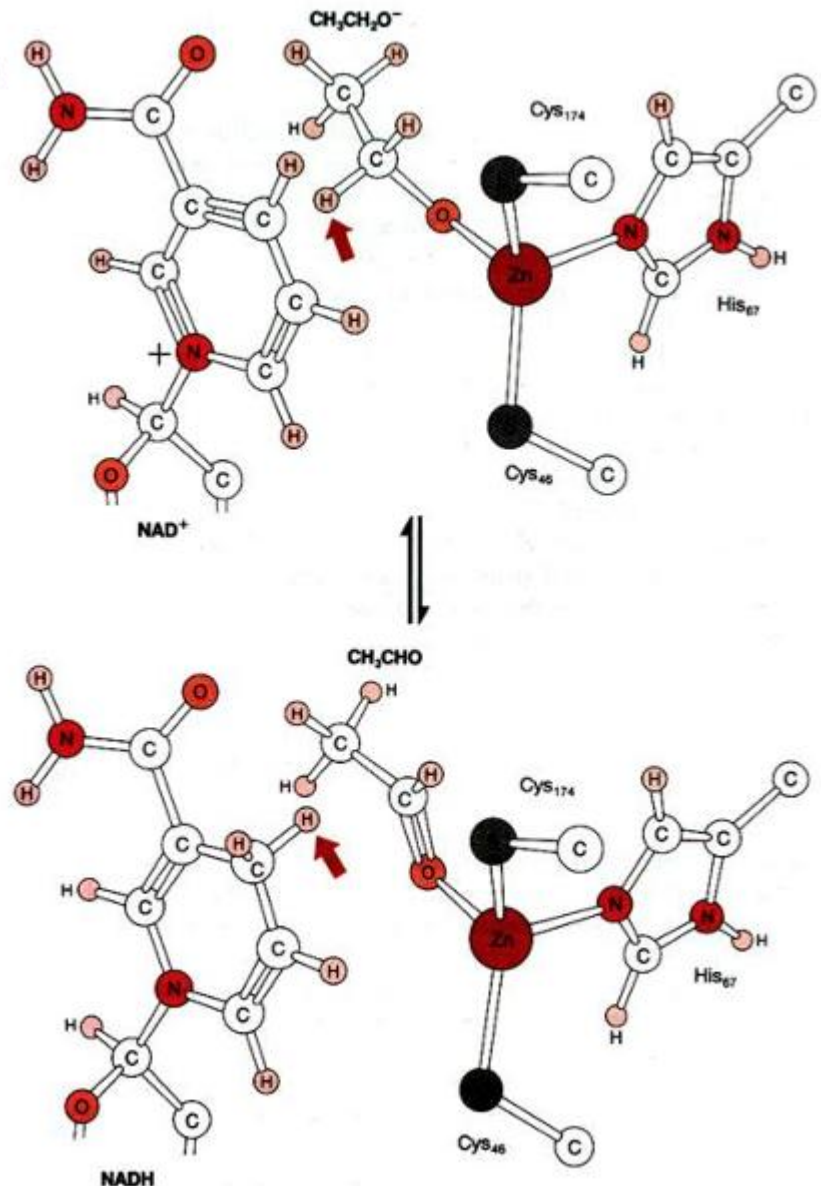
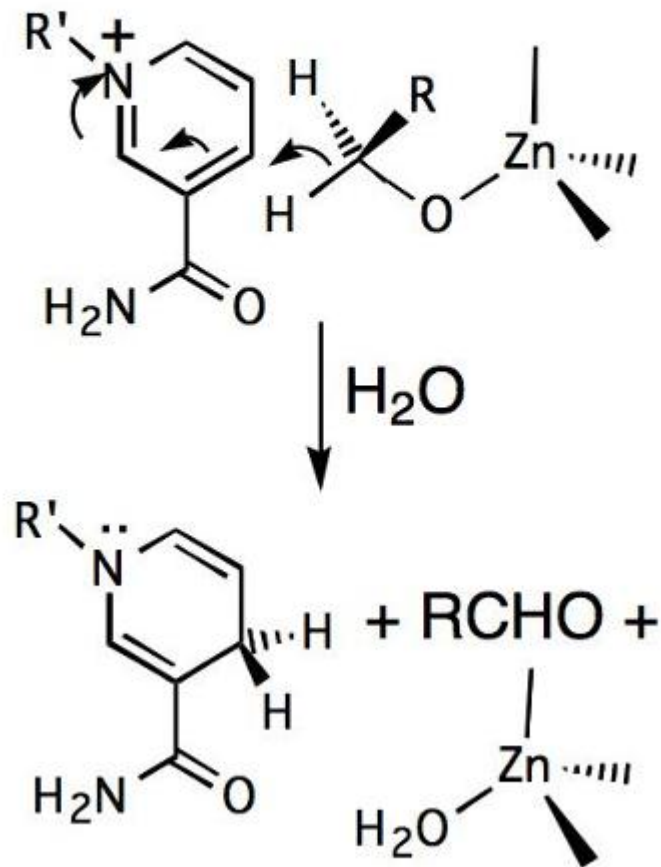
→ Peptide bonds are very stable under physiological conditions!

Liver Alcohol Dehydrogenase

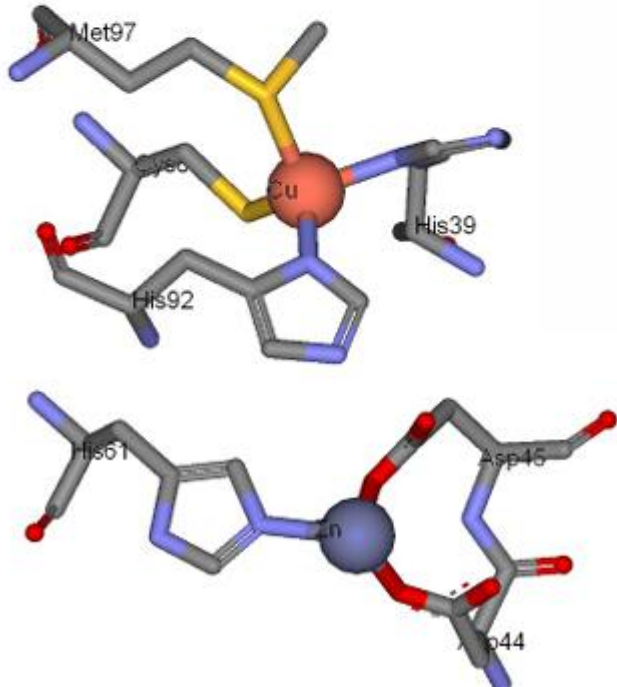
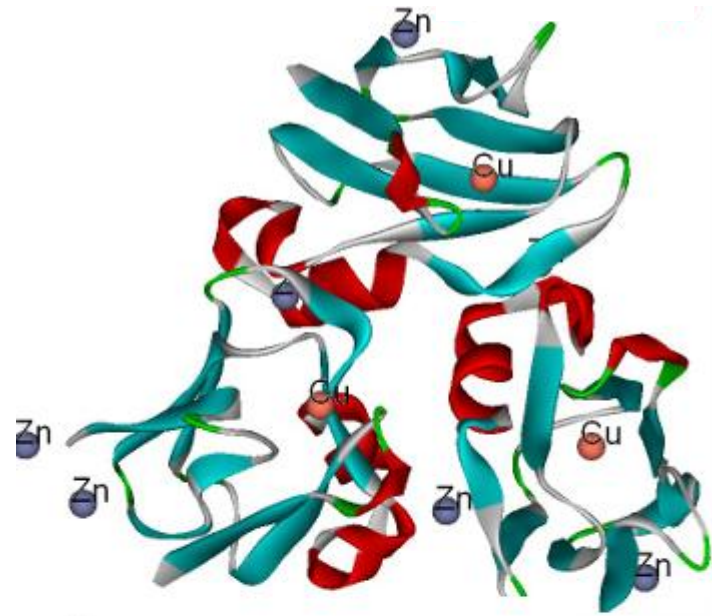
Details of hydride transfer to NAD⁺

After H⁻ transfer, aldehyde product & NADH dissociate

Hydride Transfer Mechanism



Plastocyanin



This is an electron transfer enzyme. This enzyme is able to function since Cu can undergo oxidation states of +1 and +2 easily and their inter-conversion through this protein is facile. In this enzyme, the Zn^{2+} stabilizes the protein structure that is required for the function or catalysis